

GROUP A — Short Answer Questions (5 Marks Each)

1. What is voltage? Define potential difference and write its formula.
2. Define electric current. Write its formula with the meaning of each term.
3. What is resistance? State and explain Ohm's Law.
4. What is a resistor? Write any three uses of a resistor in an electronic circuit.
5. What is drift velocity? How is it different from the random motion of electrons in a conductor?
6. Define mobility of electrons (μ_e). Write its formula and unit.
7. Write the formula for drift current and define each term used in it.
8. Differentiate between conventional flow of current and modern flow of current.
9. What is resistivity? Write its formula, unit, and symbol.
10. Define conductance. Write its formula and unit.
11. What is doping? Explain with a suitable example how it affects a semi-conductor.
12. Define Head and Penstock as used in hydroelectric power generation.
13. What is the Seebeck Effect?
14. Write short notes on: (a) Insulator (b) Conductor.

GROUP B — Long Answer Questions (8 Marks Each)

1. Explain the random motion and drift motion of electrons in a conductor with the help of diagrams. Hence derive the formula for drift velocity of electrons.
2. Explain in detail the FOUR factors affecting the resistance of a conductor (length, cross-sectional area, material, and temperature) along with their proportionality relations. Hence derive the resistivity formula $R = \rho l/A$.
3. What is nuclear energy? Explain Nuclear Fusion and Nuclear Fission with their respective reaction equations and examples.
4. Explain the working of a hydroelectric power plant with a labelled diagram, including the role of penstock, turbine, generator, and transformer. Write the EMF formula used.
5. Explain the working of a photovoltaic (solar) cell with a labelled diagram, including the terms n-type semiconductor, p-type semiconductor, and Photo-Electric Effect.

6. Classify materials on the basis of resistance into insulators, conductors, and semi-conductors. Explain each with energy band diagrams and suitable examples.
7. Differentiate between Resistance and Resistivity (any five points).

GROUP C — Numerical Problems

1. Calculate the resistance of a copper wire of length 4 m and cross-sectional area 0.015 m^2 . ($\rho_{\text{Cu}} = 1.72 \times 10^{-8} \Omega \cdot \text{m}$)
2. A silver wire has length 6 m and cross-sectional area 0.015 mm^2 . Calculate its resistance. ($\rho_{\text{Ag}} = 1.59 \times 10^{-8} \Omega \cdot \text{m}$)
3. An iron wire of length 1 km has a cross-sectional area of 0.02 cm^2 . Calculate its resistance. ($\rho_{\text{Fe}} = 10^{-7} \Omega \cdot \text{m}$)
4. A copper wire has a resistance of 25Ω and cross-sectional area 10^{-3} cm^2 . Find the length of the wire. ($\rho_{\text{Cu}} = 1.72 \times 10^{-8} \Omega \cdot \text{m}$)
5. A silver wire of length 0.8 m has a radius of 1 mm. Calculate its resistance. ($\rho_{\text{Ag}} = 1.59 \times 10^{-8} \Omega \cdot \text{m}$)
6. A wire of resistance R_1 is stretched so that its length becomes 2.5 times the original length and its cross-sectional area is reduced to half of the original. Find the new resistance in terms of R_1 .
7. A wire of resistance R is cut into four equal parts. Find the resistance of each part in terms of R .
8. A wire is stretched to 3 times its original length while its diameter is reduced to $(1/3)$ of the original. Find how many times the new resistance becomes compared to the original resistance.
9. A wire of resistance 6Ω has its length doubled and its diameter reduced to half. Find the new resistance of the wire.

*** Best of Luck! ***